

CareerNet - Data Details

Introduction and Details

CareerNet

CareerNet is a Renaissance Philanthropy-led project to develop three state-of-the-art benchmark datasets, leveraging CareerVillage.org, a platform that has crowdsourced career advice. Selecting data from over 60,000 questions and more than 3.5 million learners, the project aims to enhance AI’s ability to guide users in navigating careers and accessing social benefits.

Renaissance partnered with The Learning Agency to implement this AI benchmarking effort and the annotated datasets designed for AI model development and targeted for applications, supporting the career trajectory and upward mobility of lower-income individuals with a particular focus on reskilling, technical occupations, and allied health occupations.

Overview of Data

The repository includes 6,000 questions with 1 or more answer(s) each. It comprises three domains:

- **General career or reskilling advice** (3,000 questions)
- **Technical professions** (1,500 questions)
- **Allied Health professions (e.g., nurses, medtechs)** (1,500 questions)

The “general career advice” section offers guidance that’s relevant to any profession, while the “technical” and “health” categories dive deeper into field-specific advice. Each domain has a set of shared subdomains like job search, professional development, and reskilling. This setup keeps the datasets compatible, allowing comparisons between general and specialized responses for each career question.

During annotation, annotators flagged answers that contained bias (e.g., “Good men are needed in tech”), which each went through a second review to determine if bias was present. If so, the answer was removed.

Selecting data

To identify the subset of questions and answers that comprise the dataset (out of the 60,000 questions in total), the team followed three steps:

1. To distinguish question relevance in each domain, the team used OpenAI's GPT to parse related metadata.
 - The team identified general career advice using tags selected by the asker (i.e., “job-search”, “resume”, etc.). The team used the LLM to identify tags related to reskilling or general career navigation advice.
 - The team identified field-specific domains, technology and allied health professions, by using the LLM to categorize occupation metadata associated with each question as related to technology, health, or neither. A machine learning algorithm used by CareerVillage assigned these occupations to questions that historically achieved 80-90% accuracy.
 - The team prioritized questions related to multiple datasets for the general/reskilling dataset and the technology dataset, respectively, due to sample sizes.
2. To identify the questions, the team fine-tuned an LLM to label questions with meaningful scenario topics (more details in [Scenario Labeling](#)). These scenarios included topics such as career exploration, industry trends, credentials, job search strategies, networking, and more. The team prompted the LLM to provide 0+ topic labels for each question, such that the system did not force a topic label if one does not fit for the question, but the LLM could provide more than 1 if multiple topics are related to the question. The team derived a subset of questions (that became the curated dataset) after the labeling process so as to include a proportionate amount of questions across all topic areas.
3. Once questions were sampled based on scenarios, the team selected answers based on score (upvotes - downvotes) and recency. The team selected those with the greatest score and that were most recently provided.

After annotating these data, the team reviewed notes from the annotators to identify and remove data with biases and identify the required sample size for release.

Train/Test data splitting

The team split the data into 70/30 Train/Test sets for each domain separately such that:

- The team stratified the data by Likert Scale annotations to ensure similar distributions in Train vs Test answers.
- The team created sets ensuring no question appears in both sets.

- Goal Identification labels and Scenario labels were assessed after splitting to confirm similar distributions.

Annotation Schemes

All annotation schemes went through multiple iterations and are based on research and advisory board feedback, which consisted of experts in career navigation, Future of Work/Workforce Development, AI/ML, and NLP.

The team also conducted pilot testing on the schemes for quality assessment. Three advisory board members annotated a sample of 100 questions with approximately 3 answers each. Each coded 60 QA pairs and shared feedback on their use, suggestions, and any questions on the schemes.

Likert Scale Annotation

Human annotators rated each answer on three separate 4-point Likert scales: Completeness, Correctness, and Coherency

Completeness - How thoroughly does the answer address the question?

1 – Very Incomplete

Major ideas are missing or off-topic; the response lacks meaningful engagement with the question. Nearly every sentence is off-topic.

2 – Somewhat Incomplete

Includes some relevant points, but overall lacks detail, depth, or fails to address the full scope of the question. Many sentences are off-topic.

3 – Mostly Complete

Substantive and detailed, addressing mostly the relevant aspects of the question. Minor depth or clarity issues may be present. Most of the sentences are on topic and addressing aspects of the question.

4 – Fully Complete

Thorough, comprehensive, and coherent response. All major aspects are thoughtfully addressed. Every sentence or nearly every sentence is relevant to an aspect of the question.

Correctness - To what extent is the answer factually accurate and appropriate?

1 – Completely Incorrect

Response includes basically no correct elements. Includes many factual errors, misconceptions, or entirely inaccurate information. All or nearly all sentences contain some error or inaccuracy.

2 – Mostly Incorrect

Response is mostly incorrect. It is marred by serious inaccuracies or implausible claims. Many sentences are incorrect or contain errors.

3 – Mostly Correct

The response is mostly accurate with only minor factual errors or questionable assumptions. Most sentences are correct.

4 – Fully Correct

Entirely factually accurate and contextually appropriate; no errors present. All or nearly all sentences are correct and without issue.

Coherency - To what extent is the answer coherent and readable?

1 – Completely Incoherent

The response is extremely difficult to understand. The response lacks logical flow and clarity. All or nearly all sentences may be fragmented or disjointed, and ideas are poorly connected or entirely unrelated.

2 – Mostly Incoherent

The response has significant coherence issues. Ideas may be partially related but are poorly organized, with abrupt transitions or unclear relationships. Many sentences contain issues.

3 – Mostly Coherent

The response is generally understandable with a mostly logical structure. Some transitions may be weak, or some parts loosely connected. Most sentences are easy to follow.

4 – Fully Coherent

The response is easily understandable and is clear, well-organized, and flows smoothly. All or nearly all ideas are logically connected and easy to follow, with strong transitions and structure.

Goal Identification Annotation

Each question was labeled by human annotators according to the goal of the asker.

For each question, the annotator identified what the asker is trying to achieve by selecting one or more goal categories from the list below. Multiple goals may be selected if more than one clearly applies. Note: in data below, the numbered goals (i.e., “1. Explore Options”) are considered Overall Categories and the sub-bulleted goals (i.e. “Understand a Career Path”) are considered Detailed Categories.

1. Explore Options

The asker is trying to learn about careers, education paths, or future possibilities.

- **Understand a Career Path**

“What does a mechanical engineer do?”

- **Understand Education Options**

“What degree is best for working in finance?”

- **Explore Opportunities**

“What are some jobs you can get with strong writing skills?”

- **Explore Skills**

“What skills am I going to need to compete for current and future roles?”

2. Take Action

The asker is seeking guidance on how to do something, make a decision, or plan their future.

- **Make a Decision**

“Should I go to college right after high school or take a gap year?”

- **Get Step-by-Step Advice**

“How do I build a portfolio for graphic design?”

- **Plan a Career Path**

“What’s the best way to become a software engineer in 4 years?”

3. Understanding Purpose/Passion

The asker is trying to learn what careers, education paths or future possibilities fit their inner purpose and passions

- **Understand Intersection of Interest and Careers**

“What are some jobs that I can realistically get that meets my passion for X”

4. Get Validation/Emotional Support

The asker wants confirmation, support, or opinions to feel more confident.

- **Get Feedback or Reassurance**

“Is it bad that I haven’t figured out my major yet?”

- **Validate a Choice or Path**

“Is it smart to choose a trade instead of a degree?”

- **Seek Motivation or Encouragement**

“I feel like giving up on my goals. Any advice?”

- **Overcome Doubt or Fear**

“I’m scared I’ll fail in college. How do I stay positive?”

5. Find Resources

The asker is looking for tools, programs, or materials that will help them.

- **Find Opportunities or Programs**

“Are there any job shadowing programs for high schoolers?”

- **Get Learning Resources**

“What websites can I use to learn Excel for free?”

- **Mentors**

“Are there mentor programs in my community that offer guidance and connections to specific roles?”

- **Community**

“Are there peer communities that give me direction on certain opportunities, or help me to access wrap-around support services?”

6. Navigate Life Constraints

The asker is trying to find career or education paths that fit their personal, social, or structural circumstances. These questions are shaped by real-life barriers like financial limitations, caregiving responsibilities, transportation access, or systemic inequality.

- **Work-Life Balance / Caregiving**

“Are there any good jobs I can do from home while caring for my toddler?”

“How flexible is a nursing schedule for single parents?”

- **Financial Constraints / SES Barriers**

“I can’t afford college so what are some options that don’t require alot of money?”

“Are there high-paying jobs that don’t need a car?”

- **Geographic or Transportation Limits**

“I live in a rural area. What kinds of careers can I do remotely?”

“Are there jobs that don’t require commuting every day?”

- **Discrimination / Marginalization Awareness**

“What careers are welcoming for people with disabilities?”

“I’m undocumented, can I still go to college?”

7. Compare Options

The asker wants to weigh pros and cons between two or more options. The options must be defined.

- **Compare Roles or Paths**

“Which is better: being a nurse or a physical therapist?”

- **Compare Educational Paths**

“Is community college or a 4-year university better for computer science?”

Other Instructions

- Multi-label is allowed.
- If a question is too vague or unclear, mark it as “**Unclear Goal**”.
- If question falls outside any categories but the goal is clear mark it “**Other**”

Scenario Labeling

The team labeled each question with a Scenario Label to encompass the subcategories, career aspects being asked about, in the dataset. By labeling this first, the dataset is representative of the different kinds of questions primarily of interest to early career navigators.

A sample of 200 questions was hand-coded. An LLM selected multiple scenarios if more than one clearly applied. The team fine-tuned a GPT-4.1-mini model on 160 of the hand-coded questions. The team evaluated the fine-tuned model with the remaining 40 hand-coded questions, achieving a weighted F1 of 0.94.

- **Career Decision-Making, Transitioning, and Exploration Resources:** Questions about choosing between different career paths, making major career decisions, or about changing careers or industries, or information that helps individuals explore potential career paths.
 - Examples: How do I decide between pursuing marketing or sales? How can I move from teaching to corporate training?
- **Industry-Specific Knowledge and Trends:** Questions about particular industries, their culture, or requirements or changes and developments in specific fields
 - Examples: What does day-to-day work look like in a hospital setting? How is AI affecting job roles in marketing?
- **Skill Development:** Questions about acquiring or improving specific professional abilities/skills
 - Example: What programming languages should I learn to transition into data science?
- **Career Advancement:** Questions about moving up in a current career path
 - Example: What milestones should I aim for to advance from junior to senior engineer?
- **Non-Degree Certifications/Credentials:** Questions about professional qualifications and certifications aside from traditional college/university paths.
 - Example: Which project management certification is most valued in the healthcare industry?
- **Job Search Strategies and Infrastructure:** Questions about methods and approaches to finding job opportunities or seeking tools or platforms
 - Examples: How can I find unadvertised job openings in my field? What tools can help me determine what careers I'm suited for?
- **Resume/CV/LinkedIn Optimization and Personal Branding:** Questions about creating or improving application materials or professional image and reputation
 - Examples: How should I highlight transferable skills on my resume when changing industries? How should I position myself online to attract opportunities in my field?
- **Interview Preparation:** Questions about preparing for job interviews.
 - Example: What are common behavioral questions for UX design positions?
- **Networking and Mentorship:** Questions about building professional relationships or finding or being a mentor

- Examples: How do I follow up with someone I met at a job fair? How do I find a mentor in a field where I don't have any connections?
- **Job Offers and Negotiations:** Questions about evaluating or responding to employment offers or compensation discussions.
 - Examples: How do I compare two job offers with different benefits? When should I bring up salary during the interview process?
- **Work-Life Balance and Personal Values Alignment:** Questions about managing professional and personal life or finding work aligned with one's values
 - Examples: How do people in consulting maintain healthy boundaries with work? How can I find companies whose missions match my values and concerns?
- **Barriers, Equity, and Resource Considerations:** Questions about challenges faced by a target demographic, such as impostor syndrome and discrimination, or financial constraints.
 - Examples: What careers are welcoming for people with disabilities? What careers can be obtained with a criminal record?
- **Freelancing/Self-Employment/Entrepreneurship:** Questions about independent work.
 - Example: How do successful freelancers find clients?
- **International Careers:** Questions about working abroad or in global contexts
 - Example: What should I know before accepting a position in another country?
- **Workplace Dynamics:** Questions about navigating office politics and relationships.
 - Example: How should I handle disagreements with my direct supervisor?
- **First Job Navigation:** Questions about starting a career
 - Example: How can I stand out as a recent graduate with limited experience?
- **Mid-Career Pivots:** Questions about changing direction after establishing experience
 - Example: How can I leverage my 10 years in finance to move into sustainability?
- **Senior Career Challenges:** Questions faced by experienced professionals
 - Example: How do I stay relevant in my field after 20+ years?
- **College Queries:** Questions where individuals are seeking college or university recommendations based on their intended career path.
 - Example: What schools should I apply to if I want to be a game designer?

Standard Occupational Classification (SOC) Annotation

Human annotators assigned [Standard Occupational Classification \(SOC\)](#) codes to each question following the below instructions for every question.

After an introductory call discussing the annotation platform, scheme, instructions, and resources, annotators were provided the [list of SOC code definitions](#) and the [SOC code lookup tool](#) to aid in identifying the appropriate code with a job title or description.

Instructions:

- Read the question title and body carefully. Identify the career or careers the question is asking about.
- Add **all** SOC codes that correspond to the career(s) referenced in the question.
- If the question is clearly about **one specific detailed occupation**, select **only one** SOC code.
- If the question relates to **multiple careers**, select **all relevant codes** based on your judgment.
- Use the SOC code of the **group** that is most applicable.
- If the question does not reference any specific career or occupation, select the flag “No career mentioned in question.”
- If the question mentions a career but it doesn’t align well with any available SOC code, select the flag “Career in question not aligned with a label.” This may occur with emerging roles, very niche positions, or questions that are too vague to categorize accurately.

Reasoning Labeling

The team labeled each answer with a Reasoning Label to support LLMs’ strategic decision-making in complex, multi-step reasoning questions.

A sample of 337 answers were hand-coded. An LLM selected multiple labels if more than one clearly applied. The team fine-tuned a GPT-4.1-mini model on 269 of the hand-coded answers, addressing class imbalance by oversampling labels with less than 10 instances (2 reasoning types; each answer duplicated 3 times). The team evaluated the fine-tuned model, prompted with the rubric below, with the remaining 68 hand-coded answers, achieving a weighted F1 of 0.89.

The reasoning label rubric and labeling itself were developed in conjunction with a child reasoning expert, focusing on human reasoning present within each answer.

1. **Inductive Reasoning:** Drawing general conclusions from specific examples (e.g., the user’s past behavior). Given certain information about the specific user, then [career/education path/option] is likely/good/right for the user. E.g., “Since you’re X kind of person, you’re likely to do/like Y”. Only provide this code if the respondent explicitly

uses inductive reasoning. Statements like “this is a good job for people with [quality]” is an assertion, but it is not inductive reasoning unless it draws upon specific information about the user to inform the reasoning (e.g., the user states that they have the particular quality). If a specific intervention is suggested (e.g., “If you do X then Y will/may happen”), don’t code here; code in Predictive/Causal/Interventional. Hypotheticals like “Are you X kind of person? If so, you should Y” don’t rely on specific, known information about the user.

2. **Diagnostic/Explanatory Causal Reasoning:** Formulating plausible hypotheses that explain the data observed (e.g., X happened because of Y).
3. **Analogy about current circumstances:** Inferences from similarities between different situations or concepts; Specific example about what the respondent or other specific person(s) currently has/had in common with this user.
4. **Analogy about next steps:** Inferences from similarities between different situations or concepts; Specific example of how the respondent or other specific person(s) has/have addressed or responded to the circumstances/goals that the user has identified (This doesn’t include all reports of personal experiences in general; it includes only personal experiences that serve as examples of how the respondent or others approached what the user would like to approach; proof of concept).
5. **Predictive Causal Reasoning/Interventional Reasoning:** How actively intervening in the world would likely modify it toward some end (e.g., doing X will yield Y; “doing X will help prepare you”). Must specify some results of intervening. Include suggestions for what the user should do next given their goals/interests if the predicted outcomes of those interventions are made explicit; but do NOT code necessary conditions (code those in Necessary Conditions). Much predictive/causal reasoning is inductive; here you’re only coding things that involve how intervening will/might affect the world or self. Simply explaining that doing X will be “helpful” or will “help you” isn’t specific enough to warrant this code.
6. **Associational/Normative reasoning:** General information about what people typically do in response to what the user is facing or what the user wants to accomplish (e.g., “people who want X often do Y”). Associations, not necessarily causes. NOT specific analogies.
7. **Necessary Conditions:** What must be done to achieve an end (e.g., “you must have an advanced degree to teach at the college level”; “you must earn an MD to be a surgeon”). Often involves rules (spoken or unspoken), regulations, laws. Without X, the path/goal is impossible. Can also include explicitly mentioning what is not necessary. (Can include Deductive reasoning.) (Avoid coding colloquialisms like “you just need to be yourself”, “you need to follow your heart”).
8. **Counterfactual Reasoning:** Reasoning about hypothetical scenarios (e.g., thinking about what one’s future self might want; considering other options than the ones identified in the prompt; consider the possibility that you might end up not liking this career/opportunity that you really want to do right now).
9. **No reasoning - explanation only:** Provides information or guidance without explicitly using the other reasoning types.

Annotators and Inter-rater Reliability - Likert Scales

Training Process

For each dataset (General, Technology, Health), the team identified 2 master coders for their increased expertise in career navigation and advising in their field to identify gold standard codes in the training data and aid in aligning and guiding the annotators through the rubric and annotation. After an initial training call, the master coders guided annotators through a self-norming set (30 QA pairs; annotators received gold standard codes after annotating to review and self-reflect on codes) and 3 norming sets (150 QA pairs; after each set, IRR was run and group call attended to discuss discrepancies and align on misunderstandings).

To continue annotating post-norming, each annotator must have received a linear weighted Cohen's Kappa of 0.60+ or a 60%+ Perfect Agreement across all annotations in the final norming set (all scales included). Annotators that did not meet the requirements annotated for Goal Identification rather than continuing to annotate the Likert Scales.

For annotators that continued to the Likert Independent coding, annotators performed the remaining annotations in two sets: Double Coding and Single Coding. The Double Coding set contained 20% of the remaining QA pairs and two annotators coded each pair. Master coders adjudicated any QA pairs that received differing codes, the QA pairs randomly assigned to a single master coder for final review. After the Double Coding set, annotators and the master coders completed the Single Coding set, the remaining 80% of the QA pairs. Each of these pairs received a code from one annotator.

IRR

The below table contains the number of annotators that started in the Likert quality annotations (Initial annotator sample), the number of annotators that achieved a 0.60+ perfect agreement rate and/or 0.60+ kappa and continued to Independent coding after the norming sets (Annotators continued to Independent coding), and the number of annotators that moved to the Goal Identification labeling (Annotators to Goal Identification).

Domain	Initial annotator sample	Annotators continued to Independent coding	Annotators to Goal Identification
General	11	5	6
Health	4	3	1
Technology	5	5	0

Annotation Challenges

- Rating the quality of an answer is inherently challenging - while some answers clearly provide correct or incorrect information, others are based on personal experiences, making them harder to evaluate, especially since annotators may not have lived those experiences themselves.
- This task involves a high degree of subjectivity - how an individual defines “quality” can vary widely. Some annotators expressed that, if they had asked the question, they would have preferred more detail or a deeper explanation. This perspective made it difficult for them to separate themselves from approaching each question as if they were the asker, rather than evaluating the answer objectively as one of many possible valid responses. Consequently, many annotators struggled particularly with assessing completeness and coherency, compounded by differing tolerance levels for grammatical errors.
- Annotators brought unique life experiences and professional backgrounds to the task. Many had years of experience in career coaching but worked at different schools, with varied student populations and responsibilities. This diversity in experience influenced how each annotator judged responses, making it difficult for them to fully align their evaluations with one another.